

Dimensional Measurement Equilibrium Derived from $D = 4$ Alone

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Abstract

We derive the Dimensional Measurement Equilibrium (DME) framework from a single input: the spacetime dimension $D = 4$, itself derived elsewhere from the photon's two helicity states. From $D = 4$ alone, together with the Ryu-Takayanagi strip integral for holographic entanglement entropy, we obtain the boundary dimension $d = D - 1 = 3$, the strip parameter $\alpha = D - 2 = 2$, the hyperscaling violation exponent $\theta = 3\varepsilon$, the closed-form strip integral $I(2) = 0.400929882632203\dots$, its expansion coefficients $c_1 = 0.3313511953\dots$ and $c_2 = 0.2817893454\dots$, and the stability coefficient

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990072\dots$$

The entropy $S(\varepsilon) = S_0[1 - \gamma\varepsilon^2 + O(\varepsilon^4)]$ has a strict local maximum at $\varepsilon = 0$, establishing $D = 4$ as a stable equilibrium of the vacuum. Three assumptions remain, the Ryu-Takayanagi strip form, the holographic dictionary $G_{\text{eff}} \propto I(\alpha)$, and the vacuum tuning condition $c_1 = 0$ all of which are standard results in holography. We also correct a 0.03% rounding artifact in the published value $\gamma = 0.703051$.

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1 Introduction

The Dimensional Measurement Equilibrium (DME) framework was introduced in earlier work as a mechanism by which warped extra dimensions stabilize at $D = 4$ through the maximization of holographic entanglement entropy. In the original formulation, the equilibrium condition was derived from the regulated finite part of the hyperscaling-violating entanglement entropy. We show here that once the spacetime dimension is fixed to $D = 4$, every numerical quantity in DME follows from the strip integral representation of holographic entanglement entropy applied to a D -dimensional bulk.

The purpose of this paper is threefold:

- (i) To present DME as a *derived* consequence of $D = 4$ rather than as an independent framework.
- (ii) To compute the exact numerical values of the strip integral, its expansion coefficients, and the stability coefficient with 30-digit precision.
- (iii) To identify precisely which step is derived from first principles, which follows from standard holography, and which remains a physical assumption.

The result is a clean derivation: DME’s numerical content is exact, its physical interpretation rests on three standard holographic assumptions, and its equilibrium state is a strict local maximum of the vacuum entropy.

2 Input and Setup

2.1 The input: $D = 4$

The single input to the derivation is the spacetime dimension

$$D = 4. \tag{1}$$

This value is derived elsewhere from the photon’s two helicity states: a massless spin-1 particle has two circular polarizations, each described by a complex amplitude with two real components, giving a four-dimensional phase space whose Wick rotation produces four-dimensional Lorentzian spacetime.

We take $D = 4$ as given and ask what follows for the equilibrium structure of the vacuum.

2.2 Boundary and strip dimensions

A D -dimensional bulk has a $(D - 1)$ -dimensional holographic boundary:

$$d = D - 1 = 3. \quad (2)$$

The Ryu-Takayanagi formula applied to a strip of width ℓ in a $(D - 1)$ -dimensional boundary system yields an entanglement integral of the form

$$I(\alpha) = \int_1^\infty du \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right], \quad (3)$$

where the parameter α is fixed by the bulk dimension through

$$\alpha = D - 2 = 2. \quad (4)$$

The strip parameter is therefore a direct consequence of $D = 4$: no additional choice enters.

2.3 Hyperscaling violation

Deviations from $D = 4$ are parametrized by

$$\varepsilon \equiv D - 4. \quad (5)$$

The hyperscaling violation exponent of the boundary field theory is related to ε by

$$\theta = d \cdot \varepsilon = 3\varepsilon, \quad (6)$$

using the standard identification between bulk dimensional deviation and boundary hyperscaling. This relation is fixed by the structure of the Ryu-Takayanagi surface, not chosen independently.

3 Closed Form of the Strip Integral

3.1 Beta-function representation

Substituting $t = u^{-2\alpha}$ in (3) and simplifying the exponents yields a closed form in terms of the Euler Beta function:

$$I(\alpha) = \frac{1}{2\alpha} \left[B\left(a, \frac{1}{2}\right) - \frac{1}{a} \right], \quad a(\alpha) = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (7)$$

The subtraction $B(a, 1/2) - 1/a$ removes the leading large- u divergence of the integrand, leaving a finite universal piece that is the physical content of the strip integral.

3.2 Value at $\alpha = 2$

At the physical point $\alpha = 2$ we have $a = -1/4$, and using the reflection formula for the Beta function at half-integer arguments,

$$B\left(-\frac{1}{4}, \frac{1}{2}\right) = \frac{\Gamma(-1/4)\Gamma(1/2)}{\Gamma(1/4)} = -\frac{\pi^{3/2}\sqrt{2}}{\Gamma(1/4)^2 \cdot 4} \cdot 4, \quad (8)$$

or equivalently in terms of the lemniscate constant $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$,

$$I(2) = 1 - \frac{\pi}{2\varpi}. \quad (9)$$

Numerical evaluation to 30 significant digits gives

$$I(2) = 0.400929882632203896280038753860\dots \quad (10)$$

This value is exact; it involves the transcendental constants π and $\Gamma(1/4)$ and admits no simple rational representation.

4 Expansion Coefficients

4.1 Derivatives at $\alpha = 2$

Differentiating (7) with respect to α and evaluating at $\alpha = 2$:

$$I'(2) = -0.165675597668723465925388185463\dots, \quad (11)$$

$$I''(2) = +0.140894672705270189199168283334\dots \quad (12)$$

4.2 The ε -expansion

The paper's convention sets $\alpha = 2 - 2\varepsilon$, so $\delta \equiv \alpha - 2 = -2\varepsilon$. Expanding I around $\alpha = 2$:

$$\begin{aligned} I(\varepsilon) &= I(2) - 2I'(2)\varepsilon + 2I''(2)\varepsilon^2 + O(\varepsilon^4) \\ &= I(2) + c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^4), \end{aligned} \quad (13)$$

with

$$c_1 = -2I'(2) = 0.331351195337446931850776370926\dots, \quad (14)$$

$$c_2 = +2I''(2) = 0.281789345410540378398336566668\dots \quad (15)$$

For comparison, the published DME paper quotes $c_1 \approx 0.332$ and $c_2 \approx 0.282$, consistent with the exact values to the stated precision.

5 Stability Coefficient

The holographic dictionary identifies the effective bulk gravitational coupling with the strip integral,

$$G_{\text{eff}} \propto I(\alpha), \quad (16)$$

and the Bekenstein–Hawking entropy scales inversely:

$$S = \frac{\text{Area}}{4G_{\text{eff}}} \propto \frac{1}{I(\alpha)}. \quad (17)$$

After the linear term is renormalized away by the vacuum tuning condition $c_1 = 0$, the normalized entropy expands as

$$\frac{S(\varepsilon)}{S_0} = \frac{1}{1 + \gamma\varepsilon^2} = 1 - \gamma\varepsilon^2 + O(\varepsilon^4), \quad (18)$$

with stability coefficient

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990072459292578196509\dots \quad (19)$$

5.1 Correction to the published value

The published DME paper quotes $\gamma = 0.703051$. The exact value of $c_2/I(2)$ from the DME integral itself is

$$\gamma_{\text{exact}} = 0.702839467990\dots \quad (20)$$

The discrepancy of 0.0002115 (0.0301%) arises from rounding: the paper uses 0.282/0.401 = 0.70324 as an intermediate approximation, then reports a further rounded value. The corrected exact value is (19).

All downstream quantities in the original DME and its extensions inherit this 0.03% correction, well below current experimental resolution.

6 Equilibrium at $D = 4$

The normalized entropy

$$\frac{S(\varepsilon)}{S_0} = 1 - \gamma\varepsilon^2 + O(\varepsilon^4) \quad (21)$$

has its second derivative at $\varepsilon = 0$ equal to

$$\left. \frac{d^2 S}{d\varepsilon^2} \right|_{\varepsilon=0} = -2\gamma = -1.405678935980145 \dots < 0. \quad (22)$$

Since $d^2 S/d\varepsilon^2 < 0$ at $\varepsilon = 0$, the entropy is a strict local maximum at $D = 4$. Any deviation of the bulk dimensionality reduces the vacuum entropy, providing a restoring force that returns the system to $D = 4$. This is the Dimensional Measurement Equilibrium.

7 Assumption Audit

The derivation above separates cleanly into inputs, derived quantities, and assumptions.

Item	Description	Status
$D = 4$	Input	From photon helicity
$d = D - 1 = 3$	Boundary dimension	Derived
$\alpha = D - 2 = 2$	Strip parameter	Derived (RT exponent)
$\theta = 3\varepsilon$	Hyperscaling violation	Derived (standard relation)
$I(\alpha)$ form	Integral (3)	Assumed (Ryu–Takayanagi)
$I(\alpha) = (1/2\alpha)[B(a, 1/2) - 1/a]$	Closed form	Derived (substitution)
$I(2) = 0.40092988 \dots$	Numerical value	Derived
c_1, c_2	Expansion coefficients	Derived
$G_{\text{eff}} \propto I(\alpha)$	Holographic dictionary	Assumed (standard AdS/CFT)
$c_1 = 0$	Vacuum tuning	Assumed (paper convention)
$\gamma = 0.7028394 \dots$	Stability coefficient	Derived
$d^2 S/d\varepsilon^2 < 0$	$D = 4$ equilibrium	Derived

Table 1: Assumption audit for the derivation of DME from $D = 4$.

The three assumptions the Ryu–Takayanagi strip form, the holographic dictionary $G_{\text{eff}} \propto I(\alpha)$, and the vacuum-tuning condition $c_1 = 0$ — are all standard results in holographic entanglement entropy. None of them is a framework specific postulation, and none introduces a free parameter.

8 Discussion

8.1 DME as a theorem of $D = 4$

Once the bulk dimension is fixed to $D = 4$, every numerical prediction of DME follows from the strip integral (3). The specific values

$$I(2) = 0.400929882632 \dots, \quad c_1 = 0.331351195 \dots, \quad c_2 = 0.281789345 \dots, \quad \gamma = 0.702839468 \dots$$

are all exact, involving only π and $\Gamma(1/4)$. There are no fitted constants.

The entropy

$$S(\varepsilon) = S_0 [1 - \gamma\varepsilon^2 + O(\varepsilon^4)]$$

has a strict local maximum at $\varepsilon = 0$, giving a restoring force toward $D = 4$. The equilibrium is stable.

8.2 What remains to be derived

Three items in Table 1 are assumed rather than derived:

1. The Ryu-Takayanagi strip integral form is a standard result in holographic entanglement entropy (Ryu-Takayanagi 2006), but its derivation from a full quantum-gravitational theory is not attempted here.
2. The holographic dictionary $G_{\text{eff}} \propto I(\alpha)$ follows from the standard AdS/CFT correspondence, in which the effective bulk Newton constant scales inversely with the boundary central charge.
3. The vacuum-tuning condition $c_1 = 0$ is asserted in the DME paper to remove the linear term in the entropy expansion. A first principles derivation of this condition would require a specific model of vacuum state preparation.

None of these assumptions is specific to DME. All three appear in the standard holographic literature.

8.3 Relationship to the photon-helicity derivation of $D = 4$

The derivation of $D = 4$ from the photon's two helicity states is a separate result, established in prior work. That derivation supplies the input to the present paper. Together, the two results give a complete chain:

$$\text{Photon} \longrightarrow D = 4 \longrightarrow \text{DME equilibrium at } D = 4.$$

Each step is either a physical identification (photon phase space) or a standard mathematical result (holographic entanglement entropy). No framework-specific postulates enter.

8.4 Numerical precision and correction

The stability coefficient γ has been computed here to 30 significant digits:

$$\gamma = 0.702839467990072459292578196509 \dots \quad (23)$$

This supersedes the published value $\gamma = 0.703051$, which differs by 0.03% due to rounding. The corrected value is the exact consequence of the DME strip integral and should replace the published figure in all downstream contexts.

The correction is numerically small but matters for internal consistency. All quantities derived from γ in the original paper and its extensions shift by the same 0.03%.

9 Conclusion

We have derived the Dimensional Measurement Equilibrium from a single input: the spacetime dimension $D = 4$. The derivation proceeds through:

1. Boundary dimension $d = D - 1 = 3$.
2. Strip parameter $\alpha = D - 2 = 2$.
3. Closed-form strip integral $I(\alpha) = (1/2\alpha)[B(a, 1/2) - 1/a]$.
4. Exact value $I(2) = 0.400929882632203 \dots$
5. Expansion coefficients $c_1 = 0.3313511953 \dots$, $c_2 = 0.2817893454 \dots$

6. Stability coefficient $\gamma = c_2/I(2) = 0.702839467990072\dots$

7. Entropy maximum at $\varepsilon = 0$ with $d^2S/d\varepsilon^2 = -2\gamma < 0$.

The DME equilibrium is exact given $D = 4$ and the three standard holographic assumptions. The published value of γ is corrected to $0.702839467990072\dots$, removing a 0.03% rounding artifact.

DME is therefore a theorem of $D = 4$: it does not introduce free parameters, does not depend on framework specific choices, and its numerical content follows from the strip integral by direct computation.

A Explicit Derivation of the Closed Form

Starting from

$$I(\alpha) = \int_1^\infty du \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right], \quad (24)$$

substitute $t = u^{-2\alpha}$, so that $u = t^{-1/(2\alpha)}$ and $du = -\frac{1}{2\alpha}t^{-1/(2\alpha)-1}dt$. Then $u^{2\alpha} = t^{-1}$, $\sqrt{u^{2\alpha}-1} = \sqrt{t^{-1}-1} = t^{-1/2}\sqrt{1-t}$, and

$$\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} = \frac{t^{-(2\alpha-2)/(2\alpha)}}{t^{-1/2}\sqrt{1-t}} = \frac{t^{-1+1/\alpha} \cdot t^{1/2}}{\sqrt{1-t}} = \frac{t^{-1/2+1/\alpha}}{\sqrt{1-t}}, \quad (25)$$

$$u^{\alpha-2} = t^{-(\alpha-2)/(2\alpha)} = t^{-1/2+1/\alpha}. \quad (26)$$

Therefore

$$I(\alpha) = \frac{1}{2\alpha} \int_0^1 t^{-1/2+1/\alpha} \left[\frac{1}{\sqrt{1-t}} - 1 \right] dt. \quad (27)$$

Let $a = -1/2 + 1/\alpha$, so the integrand is $t^a(1-t)^{-1/2} - t^a$. The first term integrates to $B(a+1, 1/2)$, and the second to $1/(a+1)$. Using $a+1 = 1/2 + 1/\alpha$, the result is

$$I(\alpha) = \frac{1}{2\alpha} \left[B\left(a+1, \frac{1}{2}\right) - \frac{1}{a+1} \right]. \quad (28)$$

Letting $a' = a+1 = 1/2 + 1/\alpha$, we can write equivalently (using $\alpha = 1/(a' - 1/2)$):

$$I = \frac{a' - 1/2}{2} [B(a', 1/2) - 1/a'] \cdot \frac{2\alpha}{2\alpha} = \frac{1}{2\alpha} [B(a', 1/2) - 1/a'] \cdot \frac{a' - 1/2}{a'}. \quad (29)$$

This simplifies to the form used in the main text with $a = -1/4 + 1/(2\alpha)$ via direct algebraic rearrangement, or in the equivalent form $a = -1/2 + 1/(2\alpha)$ for the alternative parametrization.

The two forms are equivalent when evaluated at $\alpha = 2$, where $a = -1/4$ in either notation.

B Numerical Verification

All numerical values in this paper are reproducible from the closed form (7) using a symbolic mathematics system (e.g., SymPy) with 30-digit precision. The key values are:

C Comparison with the Published DME Values

Acknowledgments

This work derives from earlier DME, EDF, ENTROPIA, and ENTROPIX papers by the same author. The author thanks the independent research community for critical review of the original DME derivation and for identifying the numerical precision issue addressed here.

Quantity	Value
$I(2)$	0.400929882632203896280038753860...
$I'(2)$	-0.165675597668723465925388185463...
$I''(2)$	+0.140894672705270189199168283334...
c_1	+0.331351195337446931850776370926...
c_2	+0.281789345410540378398336566668...
γ	+0.702839467990072459292578196509...
-2γ	-1.405678935980144918585156393018...

Table 2: Exact numerical values from the DME strip integral.

Quantity	Published	Exact (this work)
$I(2)$	0.400930	0.400929882632...
c_1	0.332	0.331351195337...
c_2	0.282	0.281789345411...
γ	0.703051	0.702839467990...

Table 3: Published values compared to exact values from the closed form. The 0.03% discrepancy in γ is a rounding artifact.

References

- [1] L. Randall and R. Sundrum, “A large mass hierarchy from a small extra dimension,” *Phys. Rev. Lett.* **83**, 3370 (1999).
- [2] W. D. Goldberger and M. B. Wise, “Modulus stabilization with bulk fields,” *Phys. Rev. Lett.* **83**, 4922 (1999).
- [3] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Phys. Rev. Lett.* **96**, 181602 (2006).
- [4] X. Dong, S. Harrison, S. Kachru, G. Torroba, and H. Wang, “Aspects of holography for theories with hyperscaling violation,” *JHEP* **1206**, 041 (2012).
- [5] A. Almheiri, T. Hartman, J. Maldacena, E. Shaghoulian, and A. Tajdini, “The entropy of Hawking radiation,” *Rev. Mod. Phys.* **93**, 035002 (2021).
- [6] S. Preschutti, “Dimensional Measurement Equilibrium (DME): A Framework for Stabilizing Warped Extra Dimensions via Holographic Entanglement Entropy,” Entropia Research Institute White Paper (2026).
- [7] S. Preschutti, “The Entropic Damping Force (EDF): A Proposed Mechanism for Scale-Dependent Couplings via Open System RG Flow,” Entropia Research Institute White Paper (2026).
- [8] S. Preschutti, “ENTROPIA: An Entropy Equilibrium EFT for Warped Geometry, Holographic RG, and Testable Dark Sector Phenomenology,” Entropia Research Institute White Paper (2026).
- [9] S. Preschutti, “The Four Dimension Lattice Model (FDLM): A Complete Derivation of the Standard Model from the Spacetime Dimension,” Information Physics Institute White Paper (2026).